



Check for updates

REVIEW

REVISED Micronutrients in Adverse Pregnancy Outcomes

[version 4; peer review: 2 approved, 1 approved with reservations]

Sairoz .^{id1}, Krishnananda Prabhu^{id1}, Ranita Ghosh Dastidar^{id1},
Annayya Rao Aroor², Mahadev Rao³, Sahana shetty⁴, Vidyashree G Poojari^{id5},
Varashree BS^{id1}

¹Biochemistry, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India

²Division of Endocrinology and Metabolism, Department of Medicine, University of Missouri, Columbia, Columbia, MO, USA

³Pharmacy Practice, Manipal College of Pharmaceutical Sciences, Manipal, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India

⁴Endocrinology, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India

⁵Reproductive Medicine and Surgery, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India

v4 First published: 23 Nov 2022, 11:1369
<https://doi.org/10.12688/f1000research.124960.1>
Second version: 02 Feb 2024, 11:1369
<https://doi.org/10.12688/f1000research.124960.2>
Third version: 04 Mar 2024, 11:1369
<https://doi.org/10.12688/f1000research.124960.3>
Latest published: 21 Jun 2024, 11:1369
<https://doi.org/10.12688/f1000research.124960.4>

Abstract

About 10 to 20% of reported pregnancies have complications like spontaneous abortion (SA), preeclampsia (PE), preterm birth (PTB), and fetal growth restriction (FGR); 60% are attributed to maternal nutritional alterations. Multiple micronutrients (MMN) are supplemented in the antenatal period, but no proper validation/guidelines are available regarding dosing/time, the need for initiation, and the duration of supplementation. Studies have reported adverse pregnancy complications related to the overuse/unwanted use of multiple micronutrient supplementations during pregnancy. Identifying the exact population requiring supplementation is necessary to prevent its abuse. This article attempts to review the impacts of micronutrient deficiency/supplementation in cases of SA, FGR, and gestational diabetes mellitus (GDM), preterm delivery and PE. The study used a literature search using PubMed, Google Scholar, Mendeley, and Scopus Databases using search words pregnancy, spontaneous

Open Peer Review

Approval Status



	1	2	3
version 4			
(revision)			
21 Jun 2024			view
version 3			
(revision)			
04 Mar 2024			view
version 2			
(revision)			
02 Feb 2024			
version 1			
23 Nov 2022	view	view	

1. **Jufen Liu** ^{id}, Peking University, Beijing, China
2. **Jitendra Ahuja** ^{id}, Rajasthan University of Health Sciences, Jaipur, India

abortion, gestational diabetes mellitus (GDM), fetal growth restriction (FGR), preterm delivery, preeclampsia (PE) or “adverse pregnancy” associated with minerals, micronutrients, or supplementation. The review also considered in-house literature databases, a single-window search at Kasturba Medical College (KMC) Health sciences library, MAHE (Manipal Academy of Higher Education). The figures included in the study were created by Biorender.com. Micronutrients play multiple roles during pregnancy and fetoplacental growth stimulating growth hormone secretion, Lysyl oxidase (LOX), involved in the crosslinking between collagen and elastin in the amniotic membrane, downregulation of interleukin (IL)-1 alpha, IL-1 beta, IL-4, IL-6, IL-10, IL-12, tumor necrosis factor (TNF)-alpha and several chemokines involved in hypertension, immune-inflammatory pathways, attenuate insulin resistance, structural development of neurons and glia. Over-supplementation has led to complications such as spontaneous abortion and gestational diabetes mellitus. Since there is a lack of standardization concerning micronutrient supplementation during pregnancy, there is a need for systematic study related to the role of micronutrients during each trimester of pregnancy to optimize its supplementation and to prevent hazards associated with its abuse.

3. **Giovanni Tossetta**, Università Politecnica delle Marche, Ancona, Italy

Any reports and responses or comments on the article can be found at the end of the article.

Keywords

Micronutrients, Pregnancy, Foetal growth restriction, Gestational diabetes, Iron, spontaneous abortion, zinc, magnesium, copper



This article is included in the **Manipal Academy of Higher Education** gateway.

Corresponding author: Krishnananda Prabhu (krishnananda.prabhu@manipal.edu)

Author roles: . **S:** Conceptualization, Data Curation, Methodology, Writing – Original Draft Preparation, Writing – Review & Editing; **Prabhu K:** Conceptualization, Data Curation, Methodology, Supervision, Writing – Original Draft Preparation, Writing – Review & Editing; **Dastidar RG:** Conceptualization, Writing – Original Draft Preparation, Writing – Review & Editing; **Aroor AR:** Conceptualization, Methodology, Writing – Original Draft Preparation, Writing – Review & Editing; **Rao M:** Writing – Original Draft Preparation, Writing – Review & Editing; **shetty S:** Writing – Original Draft Preparation, Writing – Review & Editing; **Poojari VG:** Writing – Review & Editing; **BS V:** Writing – Review & Editing

Competing interests: No competing interests were disclosed.

Grant information: The author(s) declared that no grants were involved in supporting this work.

Copyright: © 2024 . S *et al.* This is an open access article distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

How to cite this article: . S, Prabhu K, Dastidar RG *et al.* **Micronutrients in Adverse Pregnancy Outcomes [version 4; peer review: 2 approved, 1 approved with reservations]** F1000Research 2024, 11:1369 <https://doi.org/10.12688/f1000research.124960.4>

First published: 23 Nov 2022, 11:1369 <https://doi.org/10.12688/f1000research.124960.1>

REVISED Amendments from Version 3

The article has been enriched with a summary Table at the end.

Any further responses from the reviewers can be found at the end of the article

Introduction

Adverse pregnancy outcomes, pregnancy other than normal livebirths, which majorly include spontaneous abortion (SA), preterm birth (PTB), fetal growth restriction (FGR), and preeclampsia (PE). 10–20% of reported pregnancies have complications like SA, PE, PTB, and FGR. Of these, 60% are attributed to maternal nutritional alterations and 40% to the alteration in the placenta and the fetus.¹ The reported adverse outcomes are 19.8% for preterm, 1.7% for spontaneous abortions, 1% for gestational diabetes, and 0.9% for stillbirths in rural India. Inadequate diet/nutrition during pregnancy commonly causes impaired fetoplacental growth and placental functions.^{2,3} As trace elements are responsible for the maintenance of cellular structures and processes, any imbalance of micronutrients such as zinc, copper, iron, and magnesium can predispose to oxidative stress and inflammation problems in conceiving, structural or functional abnormalities in the fetus, fetal death, premature ruptured membranes, and small for gestational age (SGA) babies. Inflammation, oxidative stress, Reactive Oxygen Species (ROS) generation and apoptosis can impact placental functions and pregnancy outcomes.^{4–7}

In low- and middle-income countries (LMICs) there is coexistence of micronutrient deficiency along with protein and energy deficiencies, which leads to APO, maternal depression and maternal and child cognitive impairment, premature rupture of membranes, insufficient gestational weight gain, congenital anomalies, and pre-eclampsia. But the evidence of the impact of MMS on development is limited and exposure to other environmental risk factors later in life confounds the associations.^{8–10}

Based on the above observation, multiple micronutrient supplementation (MMN) is often given during antenatal period. Antenatal micronutrients recommended for pregnant women include daily elemental iron (30–60 mg) and folic acid (0.4 mg), calcium (1.5–2 g daily), folic acid 400 µg (0.4 mg), zinc - 15 mg, selenium - 65 µg, and copper - 2 mg. As micronutrient intake and deficiencies vary across regions, countries, and populations, implementing uniform MMN supplementation during pregnancy may not be practical across the globe as the evidence is derived mainly from low-income and middle-income countries, and its applicability to high-income populations not at risk of micronutrient deficiency remains unclear.¹¹ But there are no proper validation/guidelines available concerning dosing/time, the need for initiation and duration of supplementation, etc., which has led to many unwanted supplementations. Studies report adverse pregnancy complications related to the overuse/unwanted use of multiple micronutrient supplementations during pregnancy.^{12–14} Identifying the exact population requiring supplementation is necessary to prevent its abuse. This article attempts to review the impacts of micronutrient deficiency/supplementation in cases of spontaneous abortion (SA), fetal growth restriction (FGR), preterm delivery, gestational diabetes mellitus (GDM), and preeclampsia (PE).

Methods

The study used a literature search using PubMed, Google Scholar, Mendeley, and Scopus Databases using search words “pregnancy”, OR “spontaneous abortion” OR “gestational diabetes mellitus” (GDM) OR “fetal growth restriction” (FGR) OR “preterm delivery” OR “preeclampsia” (PE) OR “adverse pregnancy” associated with “minerals,” “micronutrients,” OR “supplementation.” The review also considered in-house literature databases, a single-window search at Kasturba Medical College (KMC) Health sciences library, MAHE (Manipal Academy of Higher Education). As mentioned above, we identified 5260 articles with the search criteria (Figure 1). The study specifically screened 128 high impact journals such as The American Journal of the Medical Science, Journal of Trace Elements in Medicine and Biology, Nutrients, etc. and included 93 research articles. The review evaluated observational studies and randomized controlled trials since 1/1/2015. Meta-analyses: reviews were excluded from this study. For clinical and global importance, the review chose the referred studies.

Metabolic functions of micronutrients during pregnancy

Zinc

Zinc acts as a cofactor for more than 300 enzymes, involved in the regulation of cell growth, hormone release, organ development (kidney and heart), immunological response, maturation, and differentiation of T cells, and reproduction (ovarian development, ovarian follicular growth, oocyte maturation), transcription, synthesis of proteins, and DNA replication (Figure 2).¹⁵ Formation of Zn finger (Znf) motifs helps many transcription factors to bind DNA. Zinc is involved in bone formation, while collagen in the bone prevents bone resorption by stimulating alkaline phosphatase

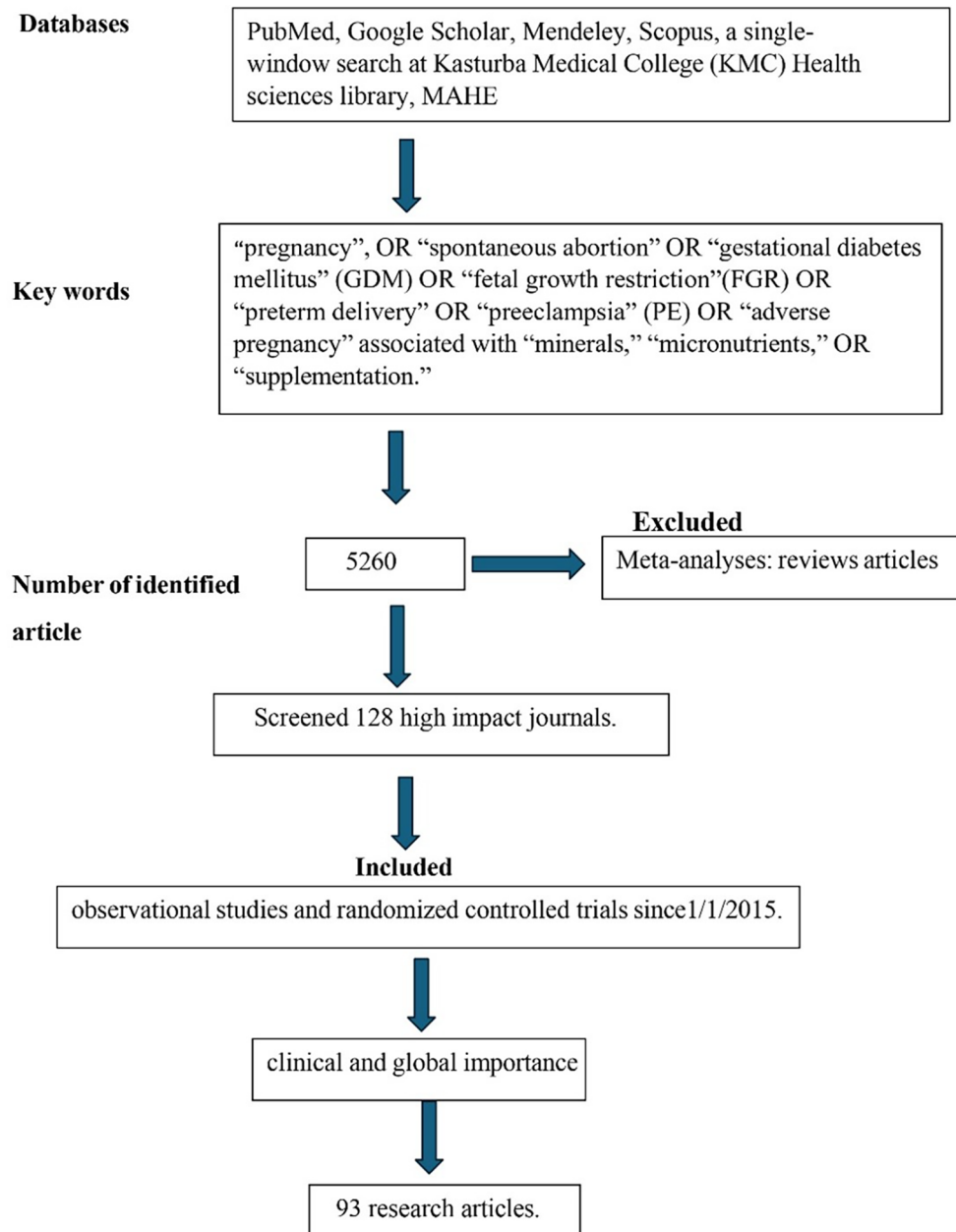


Figure 1. CONSORT diagram depicting the research methodology employed for the literature search.

(ALP) activity in the osteoblasts.¹⁶ Zinc influences many antioxidant enzymes, including Cu/Zn superoxide dismutase (SOD1), which prevents DNA damage and aids cell sensitivity towards insulin. Zinc stimulates growth promotion by stimulating growth hormone secretion, insulin like growth factor (IGF-1), insulin-like growth factor binding protein (IGFBP), and IGF-3 secretion. Insulin-like growth factor1(IGF1) promotes differentiation of the placenta, placental growth, and functional development in pregnancy during its early stages.^{16–22} Excessive zinc supplementation/zinc toxicity may lead to acute kidney injury, pancreatic injury, liver failure, and hemodynamic instability.²³

Copper

Copper is biologically essential for oxidation, energy metabolism, myelination of neurons, hematopoiesis, iron transport, and defense against free radicals.²⁰ The maternal copper level increases during pregnancy due to an increase in ceruloplasmin induced by increased estrogen. Copper in cytochromes helps in electron transport and energy generation. Copper (I) ions can produce reactive oxygen species (ROS) (Figure 3), catalyze the Haber–Weiss reaction, and induce insulin resistance by ROS leading to altered glucose homeostasis.^{15,16,21,22} Lysyl oxidase (LOX), a copper-containing

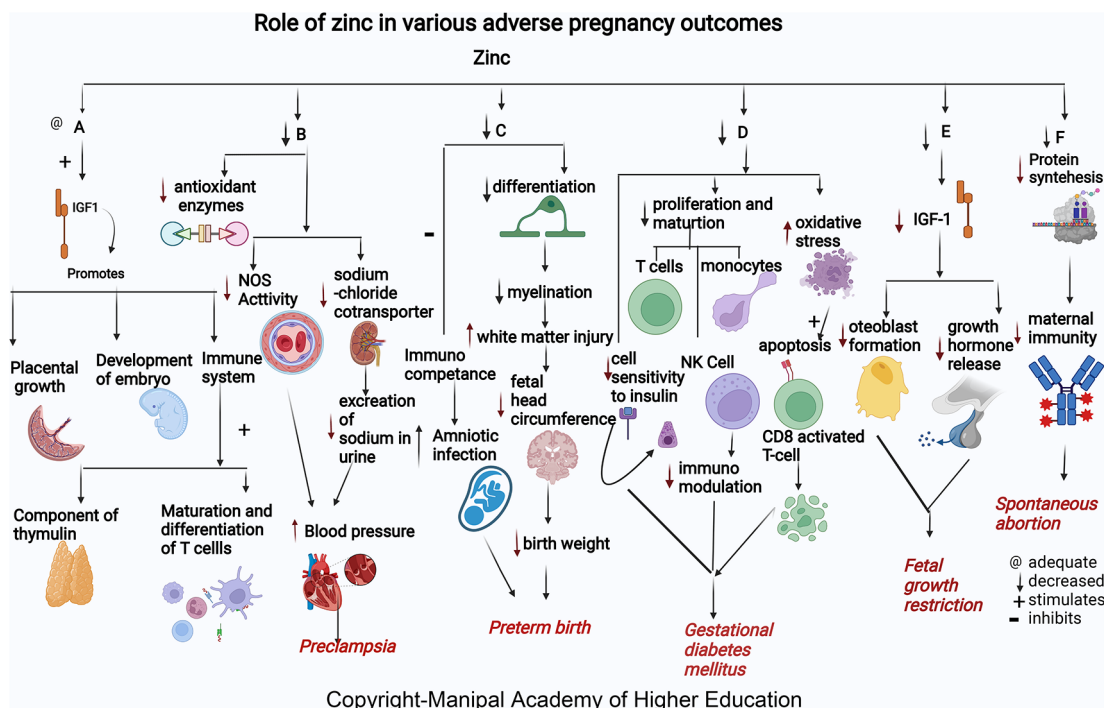


Figure 2. A-Zinc stimulates Insulin-like growth factor1 (IGF1) promotes placental growth, development of embryo and involved in immune system as component of thymulin, maturation and differentiation of T cells. B-Low zinc levels increases blood pressure by reducing the antioxidant enzymes, decreasing the nitric oxide synthase (NOS) activity in the artery and decreased effect on sodium chloride cotransporter in kidney thus reduces excretion of sodium in urine leading Preeclampsia. C-Zinc deficiency causes decreased differentiation of oligodendrocytes, decreased myelination, increased cerebral white matter injury (WMI), and decreased head circumference of the foetus causes early delivery. Lower maternal serum zinc concentration may inhibit immunological competence in both mother and foetus and therefore increase the risk of amniotic infection and lead to preterm birth. D-Zinc deficiency decreases insulin activity by decreasing cell sensitivity to insulin, decreases proliferation and maturation of T cell, NK Cell and monocytes, increases oxidative stress which induces the apoptotic deletion in the CD8+CD25+activated regulatory T cell, decreases immunomodulation and, hence leads gestational diabetes mellitus. E-Zinc deficiency decreases growth hormones and osteoblast formation via IGF-I in foetus causing foetal growth restriction. F-Zinc deficiency will decrease the protein synthesis thereby decreases the maternal immunity leading spontaneous abortion.

enzyme, is involved in oxidizing lysine and hydroxylysine residues present in precursors of elastin and collagen, also enabling crosslinking in the amniotic membrane. This crosslinking maintains mature elastin's elasticity and stabilizes collagen fibrils.²⁴ Studies conclude that elevated concentrations of Cu in the body cause brain damage due to oxidative stress, and the accumulation of Cu in the brain causes a decrease in glutathione, which is responsible for oxidative protection of neurons.²⁵

Magnesium

Magnesium is involved in >600 enzymatic reactions, such as energy metabolism, fatty acid and protein synthesis, neuromuscular excitability, and uterine hyperexcitability prevention in pregnancy, relaxation of muscles, vasodilatation and decreased vascular resistance, and transmission of nerve impulses (Figure 4). Magnesium stimulates the activation of vitamin D, thus involved in the homeostasis of calcium and phosphate, and is essential for calcium-triggered release of parathyroid hormone (PTH). Magnesium acts as an allosteric activator for adenylate cyclase, Na/K-ATPase, and phospholipase C. Magnesium downregulates IL-1 alpha, IL-4, IL-6, IL-1 beta, IL-10, IL-12, TNF-alpha and several chemokines involved in hypertension, numerous immune-inflammatory pathways, and in adverse pregnancies. Magnesium protects against pregnancy inflammation by nitric oxide synthase enzyme inhibition. It decreases proinflammatory cytokine production of activated B cell receptors by blocking the nuclear factor kappa-light-chain-enhancer.²⁶⁻²⁸ Over supplementation may lead to loss of deep tendon reflexes, sinoatrial (SA) or atrioventricular (AV) node blocks, respiratory paralysis, and, eventually, cardiac arrest.^{29,30}

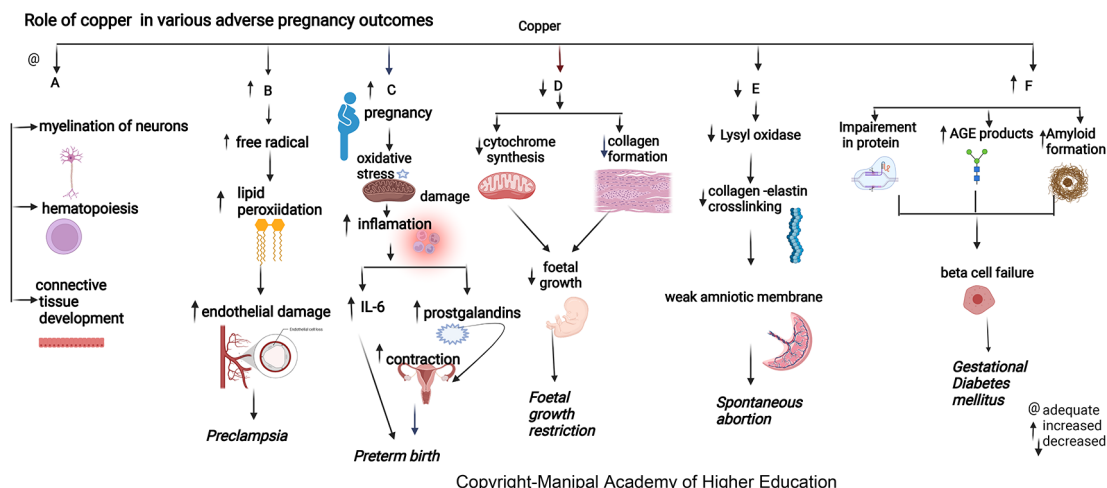


Figure 3. A-Copper is biologically important for myelination of neurons, hematopoiesis and connective tissue development. B-Increased copper contribute to disease progression by producing free radicals that lead to lipid peroxidation and damaging endothelial cell function leading preeclampsia. C-Elevated plasma copper in early pregnancy may indicate inflammation that may predispose to preterm delivery. D-Copper deficiency can hamper cytochrome and bone collagen synthesis, directly affecting fetal growth leading to Fetal growth restriction. E-Copper is involved in placentation, Lysyl oxidase's unavailability due to copper deficiency can reduce collagen and elastin crosslinking, leading to the weak amniotic membrane and spontaneous abortion. F-Copper binds nucleic acids and proteins, it plays a role in protein crosslinking, and generation of advanced glycation endproducts leading to their functional impairment.

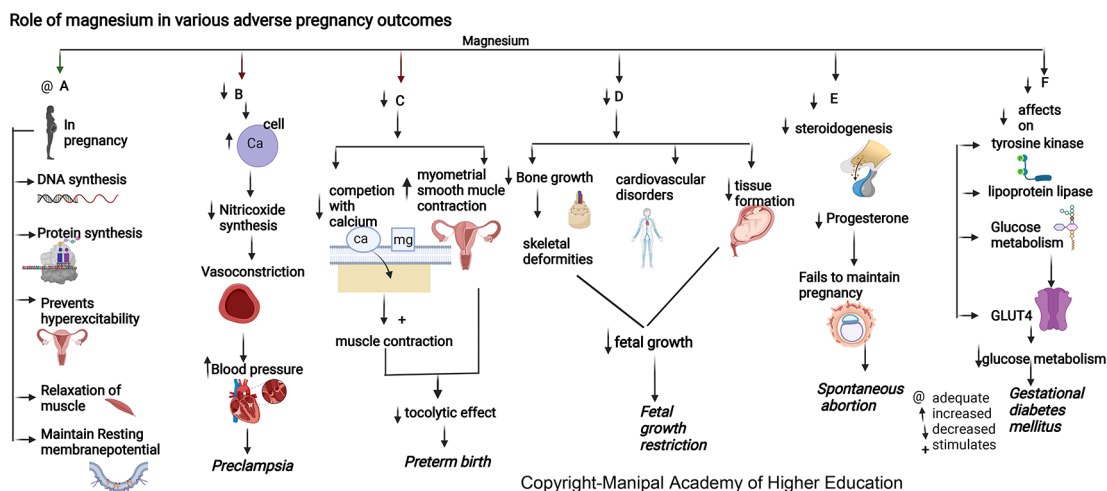


Figure 4. A-In pregnancy magnesium is involved in synthesis of DNA and proteins, prevents uterine hyperexcitability, relaxation of muscles and involved in resting membrane potential. B-Hypomagnesemia leads to increased intracellular calcium and a reduction in nitric oxide synthesis causing vasoconstriction and elevated blood pressure. C-Decreased magnesium will reduce the competition with calcium stimulating muscle contraction and increase myometrial smooth muscle contraction in uterus causing reduced tocolytic effect leading preterm birth. D-Decreased magnesium is associated with decreased bone growth, skeletal deformities, cardiovascular disorders and decreased tissue formation in foetus leading to foetal growth restriction. E-Magnesium deficiency decreases steroidogenesis (particularly progesterone synthesis) required for maintenance of pregnancy, leading to spontaneous abortion. F-Magnesium deficiency has decreased effects on tyrosine kinase, glucose transporter 4, lipoprotein lipase, decreased glucose metabolism leading Gestational diabetes mellitus.

Selenium

Selenium plays a role in antioxidative protection, protein synthesis, enzyme function, formation of thyroid hormones, and immunomodulatory and anti-proliferative mechanisms, all of which impact pregnancy. For adequate functioning of enzymes, selenium concentrations have to be about 80–95 µg/L plasma, corresponding to ~100–120 µg/L in the whole

blood. It is a vital component of the selenoproteins and glutathione peroxidase and is involved in periconceptional events such as oocyte development, fertilization, follicle growth, maturation, and implantation. Selenium, through selenoproteins like DIO2, GPX1, and selenoprotein K, plays a role in proper functioning of trophoblasts during pregnancy and an index of oxidative stress, activating signaling pathways such as phosphoinositide 3-kinase activation (PI3K/AKT) and extracellular-signal-regulated kinase (ERK). Morphological changes in the fetal cerebellum correlate with selenium levels during pregnancy. Se involves neonatal cerebellum measures, length and width, and children's motor and cognitive scores.^{21,28,31} Selenium can alleviate the inflammatory response, which attenuates insulin resistance by inhibiting the NF- κ B signaling pathway and TLR through the expression of PPAR- γ and glucose homeostasis throughout pregnancy by enhancing glucose transporters' activity (GLUT) and neutralizing reactive oxygen and nitrogen species.

Iron

Iron is integral to the transport of oxygen, energy production, cellular respiration, and DNA synthesis. Iron in ferritin prevents cellular damage and is critical to normal β cell function and glucose homeostasis (Figure 5). Iron in hemoglobin delivers oxygen through the placental unit to maintain the oxygen requirement of a fetus, of which 60% is utilized mainly for developing the nervous system to a complex sulcated and gyrated brain structure of the fetus. The homeostatic response of changes in iron protects the placenta; for example, in iron deficiency, i.e., iron import increases through transferrin receptor 1 (TFR1), and export through ferroportin (FPN) decreases.³² Excessive iron is toxic and free iron can lead to oxidative stress and inflammation. In severe cases, it can lead to ferroptosis and cell death by lipid peroxidation of cell membrane. High levels of proinflammatory cytokines in maternal serum in pregnancy maintain adequate placental, leading to inflammation and a mild systemic inflammatory response. Interleukin 6 (IL-6), the primary hepcidin regulator, disturbs iron homeostasis and leads to the expression of vascular cell adhesion molecule (VCAM1) and intercellular adhesion molecules (ICAM-1), maintaining endothelium adhesive phenotype.^{32–39} In case of over-supplementation of iron, the transferrin potential may get exhausted or exceed 85%, leading to an increase in non-transferrin-related Fe (NTBI), increasing the risk of intraocular and cardiac Fe accumulation. Excessive Fe accumulation in organs, such as the heart, liver, and endocrine glands, impairs their function, producing cardiomyopathy, cirrhosis, insulin-dependent diabetes, etc. Oxidative stress leads to faster aging of the cells, which gives a signal to the uterus to contract, resulting in premature birth.⁴⁰

Spontaneous abortion

World Health Organization (WHO) defines SA as the termination or removal of a fetus of weight ≤ 500 grams before the 20th week of gestation.^{28,37} The incidence of spontaneous miscarriage is 10–30% worldwide of clinically diagnosed

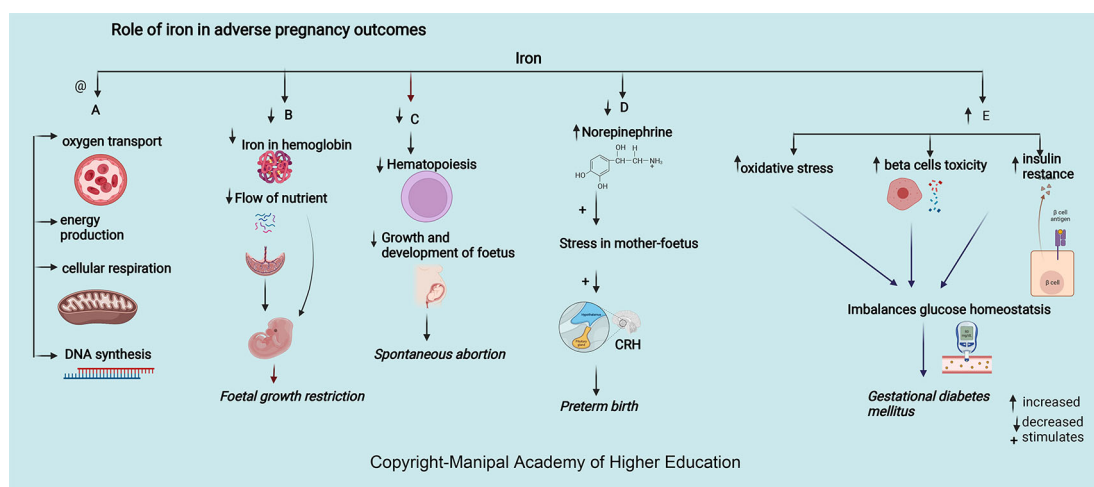


Figure 5. A-Adequate iron is involved in oxygen transport, energy production, cellular respiration, and DNA synthesis. B-Decreased Iron in haemoglobin delivers low oxygen through the maternal-placental-foetal unit leading foetal growth restriction. C-Decreased Iron leads decreased haematopoiesis, growth, and foetus development leading spontaneous abortion. D-Decreased iron increases the chances of preterm by increasing serum norepinephrine concentrations, which may induce maternal and foetal stress, stimulate the synthesis of corticotropin-releasing hormone (CRH). E-Excessive iron levels increase oxidative stress, beta cell toxicity and insulin resistance disrupting glucose homeostasis causing gestational diabetes mellitus.

pregnancies, of which 80% are within 12 weeks of pregnancy. Studies reported that the altered concentration of serum micronutrients (Zn, Cu, Se, Cd, Fe, and Mg) might lead to spontaneous abortions.³⁸

Zinc

Zinc helps in embryo development and building up of maternal immunity. Mothers who present with spontaneous abortion have significantly lower mean concentrations of serum zinc.^{20,22,38,39,41} Low serum zinc levels can lead to decreased cell generation, reduced protein synthesis, and increased cellular oxidative stress (Figure 2). Insufficient supply of antioxidants in maternal and placental circulation can predispose to complications in pregnancy.⁴² Zinc deficiency decreases antibody production, affecting maternal and fetal immunity, causing spontaneous abortion.⁴³ Many studies have shown that zinc supplementation leads to altered thymus gland secretions leading to a shift in T-helper cells. All these can predispose to recurrent spontaneous abortion in women.²²

Copper

Copper has roles in connective tissue development, hematopoiesis, energy metabolism, neurotransmitter synthesis, and myelination of neurons.⁴¹ Serum copper levels are 30% to 35% lower in women with miscarriages.³⁹ Low maternal serum copper around 15 weeks (about three and a half months) of gestation has a protective effect against SA, showing that copper is more involved in placentation than conception. In addition to other factors, optimum serum copper concentration at 15 weeks of gestation decreases spontaneous abortion.³³ Low maternal serum copper levels after 28 weeks (about six and a half months) of pregnancy were associated with higher rates of premature rupture of membranes (PROM).⁴⁴ Lysyl oxidase's unavailability due to copper deficiency can reduce collagen and elastin crosslinking, leading to a weak amniotic membrane in a spontaneous abortion case. But copper supplementations in pregnancy did not show the reversal of this effect.⁴² During pregnancy, changes in ceruloplasmin (copper-containing protein) levels can significantly change the ability of neutrophils to produce reactive oxygen species (ROS), which increases the chances of SA. As the metabolism of copper and iron is interlinked, deficiency of iron results in an elevation in liver copper. The above mechanism is linked with an elevated maternal serum copper and that of serum ceruloplasmin.⁴⁴

Selenium

Selenium, due to its antioxidant activity and involvement in the immune system, helps prevent heart diseases and cancer.^{45,46} Selenium deficiency causes decreased immune system activity, affecting both innate and adaptive immune responses. Selenium-associated enzymes such as thioredoxin reductases and glutathione peroxidases prevent reactive oxygen species-induced cell damage. Adequate selenium is responsible for improving the activation of natural killer (NK) cells and T cells. The lower levels of antioxidants due to reduced Se disturb the balance of thromboxane-prostacyclin, leading to SA. However, this has not been explored in-depth and can be studied further in the future. In pregnant women, an increased Se demand due to increased erythropoiesis in the growing fetus must be met to prevent abortion.^{26,41}

Magnesium

Magnesium (Mg) is involved in maintaining body temperature, synthesis of nucleic acid and protein (Figure 4), and maintenance of resting membrane potential in the nerve and muscle cells.⁴⁷ During pregnancy, the maternal plasma magnesium increases with gestational age. Low antenatal Mg with an increased Ca/Mg ratio may lead to spontaneous abortions.³⁸ It plays a significant role in membrane transport such as transmembrane electrolyte flux, transport of potassium and calcium ions across the plasma membrane by active transport mechanism involved in maintaining the normal functioning of vital organs such as fetal cardiac activity in the first trimester, maintenance of cell adhesion and cell migration, and neurochemical transmission involved in feto-maternal circulation. Hence, magnesium deficiency may hamper further growth and functioning of fetal organs, leading to SA. Magnesium functions in steroidogenesis (particularly progesterone synthesis) but is also involved in the maintenance of pregnancy is not clearly defined by the researchers. Studies can explore this area of the mechanism involved in the etiology of the same.^{20,41}

Cadmium

Ingested cadmium can selectively accumulate in the brain, red blood cells (RBC), kidneys, liver, and bones leading to its toxicity. This metal can displace other metals from metalloenzymes. It can seriously impact various organ morphologies and can modulate the functions of iron, calcium, copper, zinc, manganese, magnesium, etc. As living organisms poorly eliminate cadmium, it has a biological half-life of 16-30 years.⁴⁷ At 12-week pregnancy, blood cadmium >0.4 µg/L can increase the risk of embryotoxicity and spontaneous abortions.

Iron

Iron is essential in pregnancy to meet the increased demands for hematopoiesis, fetus development, and growth (Figure 5). 42% of children below five years, 50% of pregnant women-, and 33% women of reproductive age are iron deficient

worldwide.^{48,49} Among women with miscarriages, 13% higher serum iron levels are seen in 14 weeks of pregnancy. In support of the above findings, serum hepcidin and ferritin concentrations were higher in the first trimester due to possible accumulation of iron in maternal serum because the fetus is not receiving maternal iron through the placenta leading to SA.⁵⁰ Iron supplements during an antenatal period can significantly minimize the incidence of SA.⁵¹

Preterm birth

Preterm is the birth of a live baby <37 weeks of gestation.⁴¹ Reduced concentration of collagen, collagen crosslinking, increased oxidative damage biomarkers, and inflammatory signals transmitted in the placental membrane resulting from cellular apoptosis stimulate parturition in pregnancy leading to preterm delivery.⁴⁵ Copper, zinc, manganese, and selenium counterbalance the oxidative stress-induced premature aging of the placenta and inflammatory responses.⁴¹

Zinc

Approximately 18% of zinc deficiency during pregnancy causes adverse effects, leading to premature birth, increased infections, and nanism in children.⁵² Serum zinc concentration tends to decline in early pregnancy and reduces to approximately 35% less than the levels in non-pregnant women. Zinc is present as the highest content in the brain involved in migration and differentiation of the nervous system, i.e., neurogenesis (Figure 2). More than 50% of cerebral white matter injury (WMI) in premature birth is attributed to zinc deficiency.⁵³ Zinc and other micronutrients maintain the birth length (BL) and fetal birth weight (BW) and are involved in the differentiation of oligodendrocytes, which contribute to the onset of myelination. The head circumference of the fetus is the crucial factor in deciding the week of delivery.⁴³ The increased risk of amniotic infection by inhibiting maternal and fetal immunological competence due to lower maternal serum zinc concentration may lead to preterm birth. Thus, maintaining an adequate serum zinc concentration during pregnancy may help keep optimal fetal growth and prevent preterm birth.^{46,52,54}

Copper

Both copper deficiency and excess can lead to adverse pregnancy outcomes.⁵⁵ Elevated plasma copper in early pregnancy may indicate inflammation predisposing to preterm delivery (Figure 3). During pregnancy, estrogen mediates an increase in copper-carrying proteins in the blood, which may lead to a rise in circulating levels of copper. A low copper/zinc ratio reduces the risk of pregnancy complications. Studies show that lower serum copper has a protective effect early as the mature fetal liver stores the trace elements.⁵⁵ Hence, the maternal serum delivers higher copper in preterm delivery as its fetal transfer remains incomplete due to reduced pregnancy duration compared to term pregnancy.

Iron

Iron deprivation in the first and second trimesters of pregnancy can lead to premature birth and affect the baby's health.⁵⁴ During pregnancy, decreased iron and anemia increase serum norepinephrine concentrations, induce stress in mother and fetus, leading to corticotropin-releasing hormone (CRH) synthesis, and increase the chances of preterm. Increased maternal infection due to increased oxidative damage to RBC and the placental unit can result in preterm delivery.^{56,57} The requirement for iron ranges from 1–1.5 mg/day and increases to 5.0 mg/day in the last two trimesters. Due to the expansion of maternal blood volume, third-trimester pregnancy needs more iron. The risk of preterm term birth has shown an inverse correlation with the duration of iron supplementation. As iron inadequacy leads to functional impairment and excess leads to cytotoxicity, the dose of iron supplementation remains debatable; hence further studies need to focus on the amount and duration of supplementation.^{58–60}

Magnesium

Many studies reported a reduced incidence of preterm delivery with magnesium given to the mother <12 hours before delivery.^{56,57} PROM has been linked with lower serum magnesium in mothers and higher in the placenta, implying distinct pathophysiological roles for magnesium compared to preterm birth (Figure 4). Magnesium has the ability to compete with intracellular calcium in maintaining the membrane potential in preterm labor.⁶¹ Magnesium protects against premature birth by inhibiting myometrial activation and cervical maturation.^{61–63}

Fetal growth restriction

Twenty million infants globally are born as low birth per year weight babies (<2500g). Among these, approximately 95% are from developing nations. In India, fetal growth restriction is about 18 to 20%, primarily due to maternal micronutrient deficiency.^{61,62,64}

Zinc

Based on studies conducted in both animal models and humans in pregnancy, decreased maternal zinc is associated with fetal growth restrictions. Mothers are 2.6 times more likely to have FGR with low zinc levels.⁶¹ The zinc levels are significantly lower in mothers with BMI less than 18.5 who had small gestation (SGA) neonates.^{27,61,65} Maternal zinc

deficiency in early pregnancy increases the risk for FGR (Figure 2) as many enzymes and growth hormones, which play a role in fetal growth, require zinc. In FGR, osteoblast formation *via* IGF-I in the fetus gets affected due to zinc deficiency, as explained above in the metabolic functions.⁶⁶ Studies have shown that with maternal zinc supplementation, a substantial number of babies are born with appropriate birth weight, indicating that monitoring serum zinc in pregnancy is essential to ensure the proper development of the fetus and prevent FGR.⁶⁵

Copper

Copper protects cells from toxic superoxide anions to ensure average fetal growth and immune function. It also maintains the hematopoietic, central nervous, bone, and connective tissues. Maternal serum copper has an inverse relation with head circumference (HC), body weight, and the fetus's height (Figure 3).⁶³ Studies indicate that pregnant women with a deficiency of copper are prone to fetal growth restriction (FGR) as it affects collagen and elastin metabolism involved in maintaining average fetal growth (mechanism as discussed above in the metabolic section). Copper deficiency can hamper cytochrome and bone collagen synthesis, directly affecting fetal development and leading to FGR.^{61,67–70}

Iron

Pregnancy with iron deficiency reduces fetal growth (Figure 5) and affects neuronal growth, especially in the myelination of neurons, neural signals transmission, frontal cortex development, and basal ganglia.^{36,66} Decreased iron leads to decreased hemoglobin, restricting oxygen circulation and forming an environment of chronic hypoxia, leading to fetal growth restriction. Iron deficiency stimulates the synthesis of corticotropin-releasing hormone *via* increased norepinephrine and, in turn, hinders the growth of the fetus. Studies have shown that prenatal iron supplementation had a beneficial effect on baby weight in women who had taken Fe supplements during the last two trimesters of pregnancy.⁷¹ Although precise mechanisms are not clear on iron-folic acid and multiple-micronutrient supplements preventing FGR, few studies states that it maintains the average plasma volume and the flow of nutrients across the placental tissue to the fetus.^{71–73}

Magnesium

Mother and fetus tissue formation in pregnancy requires a high magnesium intake.^{62,74,75} Magnesium is involved in vitamin D activation and in maintaining calcium and phosphate physiology for bone growth. Dysregulation in either of these nutrients is associated with skeletal deformities and cardiovascular disorders, leading to fetal growth restriction. Studies have shown that magnesium supplementation to pregnant mothers can suppress the chance of intrauterine growth restriction (IUGR), optimum birth weight of baby, head circumference, and body height that can be attributed to alterations in cytokine/chemokine in the amniotic fluid and placenta.⁷⁵

Gestational Diabetes Mellitus (GDM)

Gestational diabetes mellitus is glucose intolerance at any level at the beginning of pregnancy. The high plasma fasting glucose levels may increase the chances of fetal deaths in the last weeks of gestation. Obesity and sedentary lifestyles are the main factors, and dietary intake is essential in the progression of GDM.

Zinc

Among GDM cases, there are low serum Zn levels in the second and third trimesters. Giving zinc and calcium-magnesium and vitamin D supplements for six weeks in the second trimester to subjects with GDM improves serum insulin, fasting plasma glucose levels, and the homeostatic model of measuring insulin resistance.^{76,77} Zinc intake improved fasting plasma glucose (FPG) and lipid profiles. Zinc has a role in enzymes involved in the metabolism of lipid-like action on insulin. GDM occurs *via* oxidative stress and trace elements' effects on immunomodulation and insulin activity changes. The immunomodulation mechanisms in GDM are related to oxidative stress and the impact of trace elements on the regulation of the immune system. Oxidative stress induces apoptotic deletion in the regulatory T cells, which are CD8+CD25+activated in the GDM. Besides, the Zn-insulin complex is a hexamer containing two zinc ions and forms other complexes with insulin. Therefore, zinc supplements may benefit glucose homeostasis.^{76,78}

Copper

Copper has shown positive correlations with HbA_{1c} in types 1, 2 diabetes and GDM.⁷⁹ Serum copper levels are directly involved in developing GDM independent of other established risk factors. Copper directly binds proteins affecting protein crosslinking, leading to the generation of advanced glycation end products, resulting in its functional impairment. Copper can promote progressive islet b-cell failure by causing the clustering of human amylin into an amyloid fibril. The hydrogen peroxide formed in the above process could also stimulate progressive b-cells degeneration.^{77,78,80} In contrast to the above finding, in underweight Chinese women, higher maternal serum copper has decreased the GDM risk before 24 weeks as the body composition, and body mass index (BMI) are associated with the individual trace-element status.^{78,81}

Magnesium

Maternal serum magnesium is low in the second and third trimesters with gestational diabetes mellitus.⁸² Insulin resistance might increase the placental size leading to low placental efficiency and affect fetal growth in GDM patients.⁸⁰ Magnesium, by its effects on tyrosine kinase, glucose transporter 4, and lipoprotein lipase, has a significant role in glucose metabolism (Figure 4). Supplementing magnesium and vitamin E in GDM patients resulted in a significant improvement in glycaemic control and lipid profile.^{82,83}

Iron

Maternal serum ferritin > 67.8 µg/L and iron levels 52.9 mmol/L at 12 weeks of gestation are strong predictors for developing GDM as excessive iron levels increase oxidative stress, beta cell toxicity, and insulin resistance (Figure 5), disrupting glucose homeostasis.^{77,84} Higher HbA1C levels noticed among individuals of iron deficiency have been detected.⁸⁵ Iron supplementation has not been recommended for non-anaemic pregnant women in the first and second trimesters as it could elevate post-prandial blood glucose and cause GDM.^{26,84,86,87}

Preeclampsia

Preeclampsia is hypertension in pregnancy with proteinuria of at least 0.3g/day in 2 to 8% of pregnant women and is the reason for about 63,000 deaths every year worldwide, especially in Nigeria and India.⁸⁸ Worldwide, PE management is still the leading cause for maternal and fetal mortality.^{12,19} Several micronutrients such as zinc, copper, magnesium, and calcium abnormalities play a role in preeclampsia.^{53,67,85,89}

Zinc

Many studies worldwide associated low maternal serum zinc in PE independent of geographical location. Albuminuria in PE predisposes to zinc deficiency leading to decreased delivery to cells. Reduced serum Zn concentrations in preeclampsia reflect reduced Zn binding protein and estrogen concentration.^{67,89} Low zinc levels in pregnancy can increase blood pressure by reducing the amount of calcium in the muscles and decreasing the excretion of sodium in the urine due to its effect on sodium chloride cotransporter in the kidney leading to PE.^{90,91} Zinc deficiency reduces NOS activity contributing to hypertension development by dysfunction in the endothelium, and reduces vasodilation, an endothelium-mediated activity. Zn, a signaling molecule, controls receptor and growth factor-mediated pathways in the cell's immunological mechanisms.⁹² In addition, Zinc deficiency contributes to oxidative damage by causing a negative effect on the copper-zinc superoxide dismutase enzyme system.⁹³ Lipid peroxides and antioxidants play a vital role in preeclampsia. Reduced zinc and increased Cu/Zn ratios lead to PE by inactivating antioxidant enzymes causing inadequate trophoblast invasion, suboptimal perfusion in the placenta, and ischemia.^{91,94}

Copper

In pregnant women, the lower prevalence of PE observed with high copper signifies its importance in the pathogenesis of preeclampsia.^{12,13,92} Copper levels increase under estrogenic activity, the main determinant of ceruloplasmin expression. Copper has a predictive cut-off value of 224µg/dL, indicating a good biomarker for PE diagnosis. Copper contributes to disease progression by lipid peroxidation, free radicals, and damaging endothelial cell function.^{94,95} Copper and Ceruloplasmin (cp) have a role in deciding the termination of preeclamptic pregnancies under 34 weeks (about eight months), as the copper and cp levels have shown a linear increase according to the severity of preeclampsia.⁹¹ Copper has the highest pro-oxidant capacity and serves as a cofactor to superoxide dismutase. (Figure 3).^{19,37} Since preeclampsia is a proteinopathy disorder with impaired autophagy, protein aggregates (Aβ, TTR, P-tau231, and α-syn) are generated by impaired autophagy in the placenta or kidneys.⁹⁰ In PE, the lipid peroxidation product, malondialdehyde, and altered amyloid-β production generate protein aggregates, toxic amyloid-β42 species, and their specific components α-syn, TTR, P-tau231, and Aβ.⁹⁰

Magnesium

During pregnancy, a suitable magnesium level of 360–400 mg/day and reduced serum magnesium levels can trigger preeclampsia.⁹⁴ Magnesium influences the vascular tone and contractility and plays a vital role in blood pressure regulation (Figure 4). During pregnancy, an ionized fraction of magnesium of less than 24.67% can predispose to preeclampsia. This forms the basis for using magnesium sulfate therapy for preventing and treating seizures associated with preeclampsia/eclampsia.^{81,85,86} Hypomagnesemia leads to intracellular calcium influx and a reduction in nitric oxide synthesis, causing vasoconstriction and elevated blood pressure.^{12,62,81,89} Hence Magnesium gluconate (3 g/day) has been used in managing pregnancy hypertension in high-risk pregnant women.⁹⁶

Iron

Increased iron, systemic inflammation markers, and superoxide status are associated with preeclampsia. In PE, systemic inflammation and concentrations of superoxide are higher and result in endothelial function deterioration. A 2.19-fold risk of pregnancy-induced hypertension was seen in the lowest serum iron level (≤801.20 µg/L) at 10–14 weeks

(about three months) of pregnancy as it affects trophoblast development. Reduced iron causes abnormal invasion of trophoblasts in the walls of the spiral arteries (between 6–18 weeks, about four months of pregnancy), resulting in reduced high-resistance circulation in the uteroplacental unit re-modeling, hypoxia, increased apoptosis, activation of the inflammatory and immune response. All these can lead to vascular endothelial damage and disturbance in endothelial homeostasis resulting in vasoconstriction and increased blood pressure.³⁶ In the third trimester of pregnancy, there was a positive correlation between ferritin, iron, and superoxide concentrations with vascular stiffness. The World Health Organization recommends universal iron supplementation of 30–60 mg/day throughout pregnancy. There is an increased risk of hypertension after 20 weeks of pregnancy with iron supplementation before 16-week as it causes inhibition of lymphocyte proliferation, development of oxidative stress, and elevated blood viscosity (Figure 5).^{97,98} Hence supplementation of iron along with vitamin E and antioxidant-rich foods is advisable only for pregnant women with proven iron deficiency to prevent iron overload.^{35,99,100}

Calcium

Hypocalcemia in pregnancy has seven times more risk of developing preeclampsia. A dietary deficiency in calcium may cause decreased calcium in all membrane storage sites, resulting in reduced stability in the cell membrane of vascular smooth muscle. Calcium works with magnesium in maintaining the equilibrium in the ions within the vascular membrane, thus affecting blood pressure.^{87,94,95,101} Hypocalcemia or hypovitaminosis D triggers the secretion of parathyroid hormone and renin, increasing intracellular calcium concentration and leading to vasoconstriction.^{84,93} Hence, the World Health Organization has recommended supplementation of calcium to prevent preeclampsia in pregnant women with lower intake.

Selenium

Selenium is an essential component of selenoenzymes and placental protection against oxidative stress. Selenium status is most important at 8–10 weeks (about two and a half months) gestation when there is an initiation of intervillous blood flow, posing an oxidant challenge.¹⁰² The high prevalence (56.9%) of serum selenium deficiency in the preeclamptic patient suggests that selenium deficiency is a risk factor for the development of preeclampsia and reported a serum selenium level of less than 50 µg/l (<0.633 µmol/l) in 62% of the preeclamptic patients.¹⁰³ A deficiency of selenium could lead to trophoblast cell function disorders and arouse endothelial cell activation, along with free radicals, which expedite maternal vascular malfunction, which predisposes to PE. Low levels of selenium lead to decreased glutathione peroxidases (GPx), thioredoxin reductases (Th-Redx), and selenoproteins P, antioxidative and anti-inflammatory selenoproteins needed to protect fetal, placental, and maternal tissue against oxidative stress and damage. Activating oxidative stress markers, inflammatory cytokines, fibronectin, endothelin 1, nitric oxide deficiency, prostacyclin, and vasodilators increase blood pressure.¹⁰⁴ Selenium supplementation can influence preeclampsia risk in populations with lower selenium status.¹⁰⁴

Table 1. Summary of the main results of the studies.

Adverse pregnancy outcome	Summary
Spontaneous abortion (Early pregnancy)	• Zinc deficiency predisposes to recurrent spontaneous abortion (SA) in women • Low maternal serum copper around 15 weeks of gestation has a protective effect against SA • Increased Selenium demand due to increased erythropoiesis in the growing fetus must be met to prevent abortion • Magnesium deficiency can hamper the growth and functioning of foetal organs, leading to SA
Preterm birth	• Maintaining an adequate serum zinc concentration throughout pregnancy may help keep optimal fetal growth and prevent preterm birth • Elevated plasma copper in early pregnancy may indicate inflammation predisposing to preterm delivery • Decreased maternal serum iron in early pregnancy increases the chances of preterm • Magnesium protects against premature birth in 28-40 weeks pregnancy by inhibiting myometrial activation and cervical maturation
Fetal growth restriction	• Maternal zinc deficiency in early pregnancy increases the risk for FGR • Copper deficiency in early pregnancy can hamper cytochrome and bone collagen synthesis, directly affecting fetal development and leading to FGR • Decreased serum iron throughout pregnancy leads an environment of chronic hypoxia, leading to foetal growth restriction • Decrease in magnesium in 28 weeks of pregnancy is associated with skeletal deformities and cardiovascular disorders, leading to fetal growth restriction

Table 1. *Continued*

Adverse pregnancy outcome	Summary
Gestational diabetes mellitus	- Low serum Zn levels in the second and third trimesters is associated with GDM - Higher serum copper levels in the second and third trimesters are directly involved in developing GDM independent of other established risk factors - Low maternal serum magnesium in the second and third trimesters is associated with gestational diabetes mellitus - Excessive iron levels in second trimester increase oxidative stress, beta cell toxicity, and insulin resistance leading to GDM
Preeclampsia	- Low zinc levels in pregnancy in early pregnancy can increase blood pressure leading to PE - In pregnant women, the lower prevalence of PE observed with high copper levels in second trimester - Reduced serum magnesium levels in second and third trimester can trigger preeclampsia - Increased iron, systemic inflammation markers, and superoxide status are associated with preeclampsia in late pregnancy

Conclusion

Micronutrients play multiple roles during pregnancy and fetoplacental growth like stimulating growth hormone secretion IGF-1, IGFBP, and IGF-3 secretion, leading to growth promotion, Lysyl oxidase (LOX), oxidizes lysine and hydroxylysine residues present in the precursors of elastin and collagen, enabling crosslinking in the amniotic membrane, downregulation of IL-1 alpha, IL-1 beta, IL-6, IL-4, IL-12, IL-10, TNF-alpha and several chemokines involved in hypertension, numerous immune-inflammatory pathways, fetal cerebellum morphological changes, attenuate insulin resistance and improve the activity of glucose transporters, structural development of the nervous system of the fetus, dysfunction in placenta and injury to trophoblast through hypoxia-reperfusion. Supplementation of iron along with vitamin E and antioxidant-rich foods is advisable only for pregnant women with proven iron deficiency. Iron supplementation has not been recommended for nonanemic pregnant women in the first and second trimesters as it could elevate post-prandial blood glucose and cause GDM. Calcium supplementation among with low intake can prevent preeclampsia. Zinc supplements may benefit glucose homeostasis, ensure the proper development of the fetus and prevent FGR. Pregnant women with risk factors for developing hypertension disorders could benefit from extra Mg intake. Effective implementation of community programs regarding supplementation at the regional level may contribute to the achievement of the WHO global nutrition targets and sustainable development goals. Over-supplementation had led to complications such as spontaneous abortion, and gestational diabetes mellitus. So, as there is a lack of standardization concerning micronutrient supplementation during pregnancy, there is a need for systematic study related to the role of micronutrients during each trimester of pregnancy to optimize its supplementation and to prevent hazards associated with its abuse.

Data (and Software) availability

No data are associated with this article.

Acknowledgments

The authors acknowledge Kasturba medical college, Manipal and Health sciences library, MAHE (Manipal Academy of Higher Education) for providing access to the articles reviewed in this study. The authors thank the Indian Council of Medical Research (ICMR), New Delhi, India (file no: 5/7/79/MH/Adhoc/2020-RBMCH) for funding research work to Mahadev Rao.

References

- Lane-Cordova AD, Khan SS, Grobman WA, *et al.*: **Long-Term Cardiovascular Risks Associated With Adverse Pregnancy Outcomes: JACC Review Topic of the Week.** *J. Am. Coll. Cardiol.* 2019 Apr 30; **73**(16): 2106–2116. [Publisher Full Text](#)
- Milman N, Paszkowski T, Cetin I, *et al.*: **Supplementation during pregnancy: beliefs and science.** *Gynecol. Endocrinol.* 2016 Jul 2 [cited 2022 Jan 2]; **32**(7): 509–516. [PubMed Abstract](#) | [Publisher Full Text](#)
- Kebede AS, Muche AA, Alene AG: **Factors associated with adverse pregnancy outcome in Debre Tabor town, Northwest Ethiopia: A case-control study.** *BMC. Res. Notes.* 2018 Nov 19; **11**(1): 820. [Publisher Full Text](#) | [PubMed Abstract](#) | [Free Full Text](#)
- Wang H, Hu YF, Hao JH, *et al.*: **Maternal zinc deficiency during pregnancy elevates the risks of fetal growth restriction: A population-based birth cohort study.** *Sci. Rep.* 2015 Jun 8; **5**. [Publisher Full Text](#)
- Frederiksen LE, Ernst A, Brix N, *et al.*: **Risk of adverse pregnancy outcomes at advanced maternal age.** *Obstet. Gynecol.* 2018; **131**(3): 457–463. [Publisher Full Text](#)
- Tossetta G, Fantone S, Giannubilo SR, *et al.*: **The Multifaced Actions of Curcumin in Pregnancy Outcome.** *Antioxidants (Basel).* 2021 Jan 17; **10**(1): 126. [PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)

7. Tossetta G, Fantone S, Piani F, *et al.*: **Modulation of NRF2/KEAP1 Signaling in Preeclampsia.** *Cells.* 2023 Jun 4; **12**(11): 1545.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
8. Shonibare MT, Oyinloye BE, Ajiboye BO: **Roles of Micronutrients in Pregnancy.** *Review.* 2021; **10**: 2605–2613.
9. Bourassa MW, Osendarp SJM, Adu-Afarwuah S, *et al.*: **Review of the evidence regarding the use of antenatal multiple micronutrient supplementation in low- and middle-income countries.** *Ann. N Y Acad. Sci.* 2019; **1444**: 6–21.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
10. Ahmed F: **Micronutrients and Pregnancy.** *Nutrients.* 2022; **14**: 14.
[Publisher Full Text](#)
11. Tuncalp Ö, Rogers LM, Lawrie TA, *et al.*: **WHO recommendations on antenatal nutrition: an update on multiple micronutrient supplements.** *BMJ Glob. Health.* 2020; **5**: e003375.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
12. Elmugabil A, Hamdan HZ, Elsheikh AE, *et al.*: **Serum Calcium, Magnesium, Zinc and Copper Levels in Sudanese Women with Preeclampsia.** *PLoS One.* 2016 Dec 1 [cited 2022 Jan 6]; **11**(12): e0167495.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
13. Keshavarz P, Nobakht MGBF, Mirhafez SR, *et al.*: **Alterations in Lipid Profile, Zinc and Copper Levels and Superoxide Dismutase Activities in Normal Pregnancy and Preeclampsia.** *Am. J. Med. Sci.* 2017 Jun 1 [cited 2022 Jan 6]; **353**(6): 552–558.
[PubMed Abstract](#) | [Publisher Full Text](#)
14. Okoror CEM, Enabudoso EJ, Okoror OT, *et al.*: **Serum calcium-magnesium ratio in women with preeclampsia at a tertiary hospital in Nigeria.** *Int. J. Gynaecol. Obstet.* 2020 Jun 1 [cited 2022 Jan 6]; **149**(3): 354–358.
[PubMed Abstract](#) | [Publisher Full Text](#)
15. Yücel Çelik Ö, Akdas S, Yucel A, *et al.*: **Maternal and Placental Zinc and Copper Status in Intra-Uterine Growth Restriction.** *Fetal Pediatr. Pathol.* 2020 [cited 2022 Jan 11]; **41**: 107–115.
[Reference Source](#) | [Publisher Full Text](#)
16. Alluri K, Nair KPM, Kotturu SK, *et al.*: **Transcriptional Regulation of Zinc Transporters in Human Osteogenic Sarcoma (Saos-2) Cells to Zinc Supplementation and Zinc Depletion.** *Biol. Trace Elem. Res.* 2020 Apr 1 [cited 2022 Jan 12]; **194**(2): 360–367.
[PubMed Abstract](#) | [Publisher Full Text](#)
17. Yoshida K, Urakami T, Mine Y, *et al.*: **Efficacy of zinc supplementation on growth and IGF-1 in prepubertal children with idiopathic short stature and low serum zinc levels.** *Clin. Pediatr. Endocrinol.* 2020 [cited 2022 Jan 12]; **29**(2): 63–68.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
18. Urakami T, Yoshida K, Mine Y, *et al.*: **Clinical Pediatric Endocrinology Efficacy of zinc supplementation on growth and IGF-1 in prepubertal children with idiopathic short stature and low serum zinc levels.** *Clin. Paediatr.* 2020; **29**(2): 63–68.
19. He JR, Lai YM, Liu HH, *et al.*: **Maternal IGF1, and IGF1R polymorphisms and the risk of spontaneous preterm birth.** *J. Clin. Lab. Anal.* 2017; **31**.
[Publisher Full Text](#)
20. Grieger JA, Grzeskowiak LE, Wilson RL, *et al.*: **Maternal Selenium, Copper and Zinc Concentrations in Early Pregnancy, and the Association with Fertility.** *Nutrients.* 2019 Jul 1 [cited 2022 Jan 11]; **11**(7).
[PubMed Abstract](#) | [Publisher Full Text](#)
21. Chiudzu G, Choko AT, Maluwa A, *et al.*: **Maternal Serum Concentrations of Selenium, Copper, and Zinc during Pregnancy Are Associated with Risk of Spontaneous Preterm Birth: A Case-Control Study from Malawi.** *J. Pregnancy.* 2020; **2020**: 1–7.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
22. Zare A, Saremi A, Hajhashemi M, *et al.*: **Correlation between serum zinc levels and successful immunotherapy in recurrent spontaneous abortion patients.** *J. Hum. Reprod. Sci.* 2013 Apr [cited 2022 Jan 11]; **6**(2): 147–151.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
23. Agnew UM, Slesinger TL: **Zinc Toxicity.** *StatPearls.* 2022.
24. Ogen-Shtern N, Chumin K, Cohen G, *et al.*: **Increased pro-collagen 1, elastin, and TGF-β1 expression by copper ions in an ex-vivo human skin model.** *J. Cosmet. Dermatol.* 2020 Jun 1; **19**(6): 1522–1527.
[PubMed Abstract](#) | [Publisher Full Text](#)
25. Grzeszczak K, Kwiatkowski S, Kosik-Bogacka D: **The Role of Fe, Zn, and Cu in Pregnancy.** *Biomolecules.* 2020; **10**: 1–33.
[Publisher Full Text](#)
26. Parisi F, di Bartolo I, Savasi VM, *et al.*: **Micronutrient supplementation in pregnancy: Who, what, and how much?** *Obstet. Med.* 2019 Mar 1; **12**(1): 5–13.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
27. Vázquez M, Sobrevia L, Luiza E, *et al.*: **Gestational Diabetes Mellitus: The Crosslink among Inflammation, Nitroxidative Stress, Intestinal Microbiota, and Alternative Therapies.** *Antioxidants.* 2022 Jan 7; **11**(1): 129.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
28. Parisi F, di Bartolo I, Savasi VM, *et al.*: **Micronutrient supplementation in pregnancy: Who, what, and how much?** *Obstet. Med.* 2019 Mar 1; **12**(1): 5–13.
[Publisher Full Text](#)
29. Bullarbo M, Mattson H, Broman A-K, *et al.*: **Magnesium Supplementation and Blood Pressure in Pregnancy: A Double-Blind Randomized Multicenter Study.** *J. Pregnancy.* 2018; **2018**.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
30. Ballestín SS, Campos MIG, Ballestín JB, *et al.*: **Is Supplementation with Micronutrients Still Necessary during Pregnancy? A Review.** *Nutrients.* 2021; **13**.
[Publisher Full Text](#)
31. Sangkhav F, Fisher AL, Wong S, *et al.*: **Effects of maternal iron status on placental and fetal iron homeostasis.** *J. Clin. Investig.* 2020 Feb 3; **130**(2): 625–640.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
32. Serrano NC, Guio E, Becerra-Bayona SM, *et al.*: **Scandinavian Journal of Clinical and Laboratory Investigation C-reactive protein, interleukin-6 and preeclampsia: large-scale evidence from the GenPE case-control study.**
[Publisher Full Text](#)
33. Grzeszczak K, Kwiatkowski S, Kosik-Bogacka D: **The Role of Fe, Zn, and Cu in Pregnancy.** *Biomolecules.* 2020 Aug 1 [cited 2022 Jan 11]; **10**(8): 1–33.
[PubMed Abstract](#) | [Publisher Full Text](#)
34. Zhang JY, Wang J, Lu Q, *et al.*: **Iron stores at birth in a full-term normal birth weight birth cohort with a low level of inflammation.** *Biosci. Rep.* 2020; **40**: 20202853.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
35. Lewandowska M, Sajdak S, Lubirski J: **Can Serum Iron Concentrations in Early Healthy Pregnancy Be a Risk Marker of Pregnancy-Induced Hypertension?** *Nutrients.* 2019 May 1 [cited 2022 Jan 12]; **11**(5).
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
36. Georgieff MK, Krebs NF, Cusick SE: **The Benefits and Risks of Iron Supplementation in Pregnancy and Childhood.** *Annu. Rev. Nutr.* 2019 Aug 21 [cited 2022 Jan 11]; **39**: 121–146.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
37. Krepala C, Kitporntheranunt M, Sangwipasnapanorn W, *et al.*: **Assessment of preeclampsia risk by use of serum ionized magnesium-based equation.** *Ren. Fail.* 2018 Oct 15 [cited 2022 Jan 6]; **40**(1): 99–106.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
38. Schwartz N, Sammel MD, Leite R, *et al.*: **First-trimester placental ultrasound and maternal serum markers as predictors of small-for-gestational-age infants.** *Am. J. Obstet. Gynecol.* 2014 [cited 2022 Jan 2]; **211**(3): 253.e1–253.e8.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
39. Turan K, Arslan A, Uçkan K, *et al.*: **Change of the levels of trace elements and heavy metals in threatened abortion.** *J. Chin. Med. Assoc.* 2019; **82**(7): 554–557.
[PubMed Abstract](#) | [Publisher Full Text](#)
40. Grzeszczak K, Kwiatkowski S, Kosik-Bogacka D: **The Role of Fe, Zn, and Cu in Pregnancy.** *Biomolecules.* 2020; **10**: 1176.
41. Ajayi OO, Charles-Davies MA, Arinola OG: **Progesterone, selected heavy metals, and micronutrients in pregnant Nigerian women with a history of recurrent spontaneous abortion.** *Afr. Health Sci.* 2012 [cited 2022 Jan 2]; **12**(2): 153–159.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
42. Omeljaniuk WJ, Socha K, Borawska MH, *et al.*: **Antioxidant status in women who have had a miscarriage.** *Adv. Med. Sci.* 2015 Sep 1; **60**(2): 329–334.
[Publisher Full Text](#)
43. Thaker R, Oza H, Shaikh I, *et al.*: **Correlation copper and zinc in spontaneous abortions?** *Int. J. Fertil. Steril.* 2019 Jul 1; **13**(2): 97–101.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
44. Ahmadi R, Ziaei S, Parsay S: **Association between nutritional status with spontaneous abortion.** *Int. J. Fertil. Steril.* 2017 Jan 1; **10**(4): 337–342.
[PubMed Abstract](#)
45. Rahmanian M, Sheykhnava Jahed F, Yousefi B, *et al.*: **Maternal serum copper, zinc levels and premature rupture of the fetal membranes.** *J. Pak. Med. Assoc.* 2014 Jul 14; **64**(7): 770–774. - Google Search. [cited 2022 Jan 2].
46. Kashanian M, Hadizadeh H, Faghankhani M, *et al.*: **Evaluating the effects of copper supplement during pregnancy on the premature rupture of membranes and pregnancy outcome.**

- J. Matern. Fetal Neonatal Med.* 2018 Jan 2; **31**(1): 39–46.
[Publisher Full Text](#)
47. Skarżyńska E, Zborowska H, Jakimiuk AJ, *et al.*: **Variations in serum concentrations of C-reactive protein, ceruloplasmin, lactoferrin and myeloperoxidase and their interactions during normal human pregnancy and the postpartum period.** *J. Trace Elem. Med. Biol.* 2018 Mar 1; **46**: 83–87.
[PubMed Abstract](#) | [Publisher Full Text](#)
 48. Jie O, Peng P, Qiu L, *et al.*: **Biomarkers of metal toxicity in embryos in the general population.** *J. Clin. Lab. Anal.* 2019 Oct 1; **33**(8): e22974.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 49. Saydam BK, Genc RE, Sarac F, *et al.*: **Prevalence of anemia and related factors among women in Turkey.** *Pak. J. Med. Sci.* 2017 Mar 1; **33**(2): 433–438.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 50. World Health Organization: **Nutritional anaemias: tools for effective prevention and control.** World Health Organization; 2017: pp. 1–83.
[Publisher Full Text](#)
 51. Irwinda R, Wibowo N, Putri AS: **The Concentration of Micronutrients and Heavy Metals in Maternal Serum, Placenta, and Cord Blood: A Cross-Sectional Study in Preterm Birth.** *J. Pregnancy.* 2019; **2019**: 1–7.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 52. Bakacak M, Kiliç M, Serin S, *et al.*: **Changes in Copper, Zinc, and Malondialdehyde Levels and Superoxide Dismutase Activities in Pre-Eclampsic Pregnancies.** *Medical Science Monitor: International Medical Journal of Experimental and Clinical Research.* 2015 Aug 17 [cited 2022 Jan 6]; **21**: 2414.
[Publisher Full Text](#)
 53. Rafeinia A, Tabandeh A, Khajeniāzi S, *et al.*: **Serum copper, zinc and lipid peroxidation in pregnant women with preeclampsia in gorgan.** *Open Biochem. J.* 2014 Nov 14 [cited 2022 Jan 6]; **1**(1): 83–88.
[PubMed Abstract](#) | [Publisher Full Text](#)
 54. (PDF) Maternal serum copper and zinc levels and premature rupture of the foetal membranes. [cited 2022 Jan 12].
 55. Skarżyńska E, Zborowska H, Jakimiuk AJ, *et al.*: **Variations in serum concentrations of C-reactive protein, ceruloplasmin, lactoferrin and myeloperoxidase and their interactions during normal human pregnancy and postpartum period.** *J. Trace Elem. Med. Biol.* 2018 Mar 1 [cited 2022 Jan 12]; **46**: 83–87.
[PubMed Abstract](#) | [Publisher Full Text](#)
 56. Ramakrishnan U, Nguyen PH, Gonzalez-Casanova I, *et al.*: **Neither preconceptional weekly multiple micronutrient nor iron-folic acid supplements affect birth size and gestational age compared with a folic acid supplement alone in rural Vietnamese women: a randomized controlled trial.** *J. Nutr.* 2016 Jul 1; **146**(7): 1445S–1452S.
[PubMed Abstract](#) | [Publisher Full Text](#)
 57. Kang Y, Dang S, Zeng L, *et al.*: **Multi-micronutrient supplementation during pregnancy for prevention of maternal anaemia and adverse birth outcomes in a high-altitude area: a prospective cohort study in rural Tibet of China.** *Br. J. Nutr.* 2017 Sep; **118**(6): 431–440. - Google Search. [cited 2022 Jan 2].
[PubMed Abstract](#) | [Publisher Full Text](#)
 58. Godswill AG, Somtochukwu IV, Ikechukwu AO, *et al.*: **Health Benefits of Micronutrients (Vitamins and Minerals) and their Associated Deficiency Diseases: A Systematic Review.** *Int. J. Food Sci.* 2020 Jan 7 [cited 2022 Jan 2]; **3**(1): 1–32.
[Publisher Full Text](#) | [Reference Source](#)
 59. Rees WD, Hay SM, Hayes HE, *et al.*: **Iron deficiency during pregnancy and lactation modifies the fatty acid composition of the brain of neonatal rats.** *J. Dev. Orig. Health Dis.* 2020 Jun; **11**(3): 264–272. - Google Search. [cited 2022 Jan 2].
[Publisher Full Text](#)
 60. Shao Y, Mao B, Qiu J, *et al.*: **Association between iron supplementation, dietary iron intake and risk of moderate preterm birth: A birth cohort study in china.** *Iran. J. Public Health.* 2021 [cited 2022 Feb 4]; **50**(6): 1177–1187.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 61. Kucukaydin Z, Kurdoglu M, Kurdoglu Z, *et al.*: **Selected maternal, fetal and placental trace element and heavy metal and maternal vitamin levels in preterm deliveries with or without preterm premature rupture of membranes.** *J. Obstet. Gynaecol. Res.* 2018 May; **44**(5): 880–889. - Google Search. [cited 2022 Jan 2].
[Publisher Full Text](#)
 62. Uwitonze AM, Razzaque MS: **Role of Magnesium in Vitamin D Activation and Function.** *J. Am. Osteopath. Assoc.* 2018 Mar 1 [cited 2022 Jan 4]; **118**(3): 181–189.
[Publisher Full Text](#)
 63. Fanni D, Gerosa C, Nurchi VM, *et al.*: **The Role of Magnesium in Pregnancy and in Fetal Programming of Adult Diseases.** *Biol. Trace Elem. Res.* 2021 Oct 1 [cited 2022 Jan 11]; **199**(10): 3647–3657.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 64. Turitz AL, Too GT, Gyamfi-Bannerman C: **Proximity of magnesium exposure to delivery and neonatal outcomes.** *Am. J. Obstet. Gynecol.* 2016 Oct 1; **215**(4): 508.e1–508.e6.
[Publisher Full Text](#)
 65. Li Z, Liang C, Huang K, *et al.*: **Umbilical serum copper status and neonatal birth outcomes: a prospective cohort study.** *Biol. Trace Elem. Res.* 2018 Jun; **183**(2): 200–208. - Google Search. [cited 2022 Jan 2].
[PubMed Abstract](#) | [Publisher Full Text](#)
 66. Jyotsna S, Amit A, Kumar A: **Study of Serum Zinc in Low Birth Weight Neonates and Its Relation with Maternal Zinc.** *J. Clin. Diagn. Res.* 2015 Jan 1 [cited 2022 Jan 4]; **9**(1): SC01–SC03.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 67. Sarwar MS, Ahmed S, Ullah MS, *et al.*: **Comparative study of serum zinc, copper, manganese, and iron in preeclamptic pregnant women.** *Biol. Trace Elem. Res.* 2013 Jul [cited 2022 Jan 6]; **154**(1): 14–20.
[PubMed Abstract](#) | [Publisher Full Text](#)
 68. Zhao H, Wu W, Zhang X, *et al.*: **Associations between molybdenum exposure and ultrasound measures of fetal growth parameters: A pilot study.** *Chemosphere.* 2021 Apr 1; **269**: 128709.
[PubMed Abstract](#) | [Publisher Full Text](#)
 69. Sabra S, Malmqvist E, Saborit A, *et al.*: **Heavy metals exposure levels and their Correlation with different clinical forms of fetal growth restriction.** *PLoS One.* 2017 Oct 1 [cited 2022 Jan 4]; **12**(10): e0185645.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 70. Kojima C, Shoji H, Ikeda N, *et al.*: **Association of zinc and copper with clinical parameters in the preterm newborn.** *Pediatr. Int.* 2017 Nov 1; **59**(11): 1165–1168.
[PubMed Abstract](#) | [Publisher Full Text](#)
 71. Flynn AC, Begum S, White SL, *et al.*: **Relationships between Maternal Obesity and Maternal and Neonatal Iron Status.** *Nutrients.* 2018 Aug 1 [cited 2022 Jan 6]; **10**(8).
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 72. Vohr BR, Davis EP, Wanke CA, *et al.*: **Neurodevelopment: The impact of nutrition and inflammation during preconception and pregnancy in low-resource settings.** *Pediatrics.* 2017 Apr 1; **139**: S38–S49.
[PubMed Abstract](#) | [Publisher Full Text](#)
 73. Yang J, Cheng Y, Pei L, *et al.*: **Maternal iron intake during pregnancy and birth outcomes: a cross-sectional study in Northwest China.** *Br. J. Nutr.* 2017 Mar 28 [cited 2022 Jan 4]; **117**(6): 862–871.
[Publisher Full Text](#)
 74. Vani Axita C, Stephen S, Nadasha T: **Estimation of serum levels of magnesium in antenatal women in a tertiary health centre.** *Int. J. Clin. Biochem. Res.* 2021 Oct 28 [cited 2022 Feb 7]; **8**(3): 204–210.
[Publisher Full Text](#) | [Reference Source](#)
 75. Zarean E, Tarjan A: **Effect of Magnesium Supplement on Pregnancy Outcomes: A Randomized Control Trial.** *Adv. Biomed. Res.* 2017 [cited 2022 Jan 2]; **6**(1): 109.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 76. Karamali M, Bahramimoghdam S, Sharifzadeh F, *et al.*: **Magnesium-zinc-calcium-vitamin d co-supplementation improves glycemic control and markers of cardiometabolic risk in gestational diabetes: A randomized, double-blind, placebo-controlled trial.** *Appl. Physiol. Nutr. Metab.* 2018; **43**(6): 565–570.
[PubMed Abstract](#) | [Publisher Full Text](#)
 77. Looman M, Geelen A, Samlal RA, *et al.*: **Changes in micronutrient intake and status, diet quality and glucose tolerance from preconception to the second trimester of pregnancy.** *Nutrients.* 2019 Feb; **11**(2): 460. - Google Search. [cited 2022 Jan 5].
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
 78. Mishu FA, Boral N, Ferdous N, *et al.*: **Estimation of Serum Zinc, Copper and Magnesium Levels in Bangladeshi women with Gestational Diabetes Mellitus Attending in a Tertiary Care Hospital.** *Mymensingh Med. J.: MMJ.* 2019 Jan 1; **28**(1): 157–162. - Google Search. [cited 2022 Jan 5].
[PubMed Abstract](#) | [Publisher Full Text](#)
 79. Li P, Yin J, Zhu Y, *et al.*: **Association between plasma concentration of copper and gestational diabetes mellitus.** *Clin. Nutr.* 2019 Dec 1 [cited 2022 Jan 7]; **38**(6): 2922–2927.
[PubMed Abstract](#) | [Publisher Full Text](#)
 80. Tanaka K, Yamada K, Matsushima M, *et al.*: **Increased maternal insulin resistance promotes placental growth and decreases placental efficiency in pregnancies with obesity and gestational diabetes mellitus.** *J. Obstet. Gynaecol. Res.* 2018 Jan 1; **44**(1): 74–80.
[PubMed Abstract](#) | [Publisher Full Text](#)

81. Prospective association of metal levels with gestational diabetes mellitus and glucose: A retrospective cohort study from South China | Elsevier Enhanced Reader. [cited 2022 Jan 5].
82. Bullarbo M, Mattson H, Broman AK, *et al.*: **Magnesium Supplementation and Blood Pressure in Pregnancy: A Double-Blind Randomized Multicenter Study.** *J. Pregnancy.* 2018; **2018**: 1–10.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
83. Maktabi M, Jamilian M, Amirani E, *et al.*: **The effects of magnesium and vitamin E co-supplementation on parameters of glucose homeostasis and lipid profiles in patients with gestational diabetes.** *Lipids Health Dis.* 2018 Dec; **17**(1): 1–6. - Google Search. [cited 2022 Jan 5]
[Publisher Full Text](#)
84. Guo W, Wang H, Liu Q, *et al.*: **Analysis of the Correlation of gestational diabetes mellitus and peripheral ferritin with iron levels in early pregnancy.** *Minerva Endocrinol.* 2018 Feb 13; **44**(1): 91–96. - Google Search. [cited 2022 Jan 5].
[PubMed Abstract](#) | [Publisher Full Text](#)
85. Zhu B, Liang C, Xia X, *et al.*: **Iron-Related Factors in Early Pregnancy and Subsequent Risk of Gestational Diabetes Mellitus: the Ma'anshan Birth Cohort (MABC) Study.** *Biol. Trace Elem. Res.* 2019 Sep 15 [cited 2022 Feb 11]; **191**(1): 45–53.
[PubMed Abstract](#) | [Publisher Full Text](#)
86. Zhu B, Liang C, Xia X, *et al.*: **Iron-related factors in early pregnancy and subsequent risk of gestational diabetes mellitus: the Ma'anshan birth cohort (MABC) study.** *Biol. Trace Elem. Res.* 2019 Sep; **191**(1): 45–53. - Google Search. [cited 2022 Jan 5].
[PubMed Abstract](#) | [Publisher Full Text](#)
87. Si S, Shen Y, Xin X, *et al.*: **Hemoglobin concentration and iron supplement during pregnancy were associated with an increased risk of gestational diabetes mellitus.** *J. Diabetes.* 2021 Mar 1; **13**(3): 211–221.
[PubMed Abstract](#) | [Publisher Full Text](#)
88. Okoror CEM, Enabudoso EJ, Okoror OT, *et al.*: **Serum calcium-magnesium ratio in women with preeclampsia at a tertiary hospital in Nigeria.** *Int. J. Gynaecol. Obstet.* 2020 Jun 1 [cited 2022 Jan 12]; **149**(3): 354–358.
[PubMed Abstract](#) | [Publisher Full Text](#)
89. Enebe JT, Dim CC, Ugwu EO, *et al.*: **Serum antioxidant micronutrient levels in preeclamptic pregnant women in Enugu, south-East Nigeria: A comparative cross-sectional analytical study.** *BMC Pregnancy Childbirth.* 2020 Jul 6 [cited 2022 Jan 6]; **20**(1): 1–9.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
90. Cheng S, Banerjee S, Daiello LA, *et al.*: **Novel blood test for early biomarkers of preeclampsia and Alzheimer's disease.** *Sci. Rep.* 2021 Dec 1; **11**(1)
[Publisher Full Text](#)
91. Sak S, Barut M, Çelik H, *et al.*: **Copper and ceruloplasmin levels are closely related to the severity of preeclampsia.** *J. Matern. Fetal Neonatal Med.* 2020 Jan 2 [cited 2022 Jan 6]; **33**(1): 96–102.
[PubMed Abstract](#) | [Publisher Full Text](#)
92. Bakacak MB, Kılınç MB, Serin SD, *et al.*: **Changes in Copper, Zinc, and Malondialdehyde Levels and Superoxide Dismutase Activities in Preeclamptic Pregnancies.** 2015.
93. Mihatsch W, Fewtrell M, Goulet O, *et al.*: **ESPGHAN/ESPEN/ESPR/CSPEN guidelines on pediatric parenteral nutrition: Calcium, phosphorus and magnesium.** *Clin. Nutr.* 2018 Dec 1 [cited 2022 Jan 6]; **37**(6 Pt B): 2360–2365.
[PubMed Abstract](#) | [Publisher Full Text](#)
94. Darkwa EO, Antwi-Boasiako C, Djagbletey R, *et al.*: **Serum magnesium and calcium in preeclampsia: a comparative study at the Korle-Bu Teaching Hospital, Ghana.** *Integrated Blood Pressure Control.* 2017 Aug 16 [cited 2022 Jan 6]; **10**: 9–15.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
95. Al-Jameil N, Tabassum H, Ali MN, *et al.*: **Correlation between serum trace elements and risk of preeclampsia: A case controlled study in Riyadh, Saudi Arabia.** *Saudi J. Biol. Sci.* 2017 [cited 2022 Jan 6]; **24**(6): 1142–1148.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
96. de Araújo CAL, de Sousa OL, de Gusmão IMB, *et al.*: **Magnesium supplementation and preeclampsia in low-income pregnant women - A randomized double-blind clinical trial.** *BMC Pregnancy Childb.* 2020 Apr 9 [cited 2022 Jan 6]; **20**(1): 1–6.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
97. Yousry MA, Radwan AM, Gebreel MA, *et al.*: **Prevalence of Maternal Anemia in Pregnancy: The Effect of Maternal Hemoglobin Level on Pregnancy and Neonatal Outcome.** *Open J. Obstet. Gynecol.* 2018 Jun 26 [cited 2022 Jan 6]; **08**(7): 676–687.
[Publisher Full Text](#)
98. Jirakittidul P, Sirichotiyakul S, Ruengorn C, *et al.*: **Effect of iron supplementation during early pregnancy on the development of gestational hypertension and preeclampsia.** *Arch. Gynecol. Obstet.* 2018 Sep 1 [cited 2022 Jan 6]; **298**(3): 545–550.
[PubMed Abstract](#) | [Publisher Full Text](#)
99. Mannaerts D, Faes E, Cos P, *et al.*: **Oxidative stress in healthy pregnancy and preeclampsia is linked to chronic inflammation, iron status and vascular function.** *PLoS One.* 2018 Sep 1 [cited 2022 Jan 6]; **13**(9): e0202919.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
100. Birhanu Z, Chapleau GM, Ortolano SE, *et al.*: **Ethiopian women's perspectives on antenatal care and iron-folic acid supplementation: Insights for translating global antenatal calcium guidelines into practice.** *Matern. Child Nutr.* 2018 Feb 1 [cited 2022 Jan 6]; **14**(Suppl 1): e12424.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
101. Abbasalizadeh S, Abam F, Mirghafourvand M, *et al.*: **Comparing levels of vitamin D, calcium and phosphorus in normotensive pregnant women and pregnant women with preeclampsia.** *J. Obstet. Gynaecol.* 2020 Nov 16 [cited 2022 Jan 6]; **40**(8): 1069–1073.
[PubMed Abstract](#) | [Publisher Full Text](#)
102. Holmquist E, Brantsæter AL, Meltzer HM, *et al.*: **Maternal selenium intake and selenium status during pregnancy in relation to preeclampsia and pregnancy-induced hypertension in a large Norwegian Pregnancy Cohort Study.** *Sci. Total Environ.* 2021 Dec 1; **798**: 149271.
[PubMed Abstract](#) | [Publisher Full Text](#)
103. Eze SC, Ododo NA, Ugwu EO, *et al.*: **Serum selenium levels of preeclamptic and normal pregnant women in Nigeria: A comparative study.** *PLoS One.* 2020 Aug 1 [cited 2022 Jan 6]; **15**(8): e0238263.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)
104. Lewandowska M, Więcekowska B, Sajdak S, *et al.*: **First Trimester Microelements and Their Relationships with Pregnancy Outcomes and Complications.** *Nutrients.* 2020 Apr 1; **12**(4): 1108.
[PubMed Abstract](#) | [Publisher Full Text](#) | [Free Full Text](#)

Open Peer Review

Current Peer Review Status:



Version 4

Reviewer Report 24 June 2024

<https://doi.org/10.5256/f1000research.167997.r293774>

© 2024 Tossetta G. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Giovanni Tossetta

Università Politecnica delle Marche, Ancona, Italy

The manuscript has been significantly improved and can be accepted in the present form

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Pregnancy complications

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Version 3

Reviewer Report 25 May 2024

<https://doi.org/10.5256/f1000research.162687.r281711>

© 2024 Tossetta G. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Giovanni Tossetta

¹ Università Politecnica delle Marche, Ancona, Italy

² Università Politecnica delle Marche, Ancona, Italy

the manuscript is interesting and generally well written and illustrated. Figures are clear and easily understandable. Moreover, manuscript has been significantly improved after revision. However, two points could be improved.

Introduction: this section could be improved since it is too superficial for a review article. In fact, it could be highlighted that oxidative stress and inflammation are two common characteristics of the pregnancy complications treated in this manuscript (see for example PMID: 33477354, PMID: 37296665).

A table summarizing the main results of the studies treated would be helpful

References

1. Tossetta G, Fantone S, Piani F, Crescimanno C, et al.: Modulation of NRF2/KEAP1 Signaling in Preeclampsia. *Cells*. 2023; **12** (11). [PubMed Abstract](#) | [Publisher Full Text](#)
2. Tossetta G, Fantone S, Giannubilo SR, Marzioni D: The Multifaced Actions of Curcumin in Pregnancy Outcome. *Antioxidants (Basel)*. 2021; **10** (1). [PubMed Abstract](#) | [Publisher Full Text](#)

Is the topic of the review discussed comprehensively in the context of the current literature?

Yes

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Yes

Are the conclusions drawn appropriate in the context of the current research literature?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Pregnancy complications

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Author Response 27 May 2024

Krishnananda Prabhu

Dear Sir/Madam,

Thank you very much for your valuable inputs in enriching this article. We shall incorporate the suggestions soon and resend

Competing Interests: No competing interests were disclosed.

Version 1

Reviewer Report 19 January 2024

<https://doi.org/10.5256/f1000research.137211.r199841>

© 2024 Ahuja J. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Jitendra Ahuja 

¹ Rajasthan University of Health Sciences, Jaipur, Rajasthan, India

² Rajasthan University of Health Sciences, Jaipur, Rajasthan, India

Relevance and Description:

The article addresses a pertinent and current issue as adverse pregnancy outcomes remain a significant concern in the realm of maternal and fetal health. Micronutrient supplementation during pregnancy is a common practice, but the absence of standardized guidelines has raised concerns about potential overuse and its consequent adverse effects. Grasping the role of micronutrients during pregnancy and their connection to complications is pivotal for enhancing the well-being of both mother and fetus.

Major Points:

Clear Problem Statement: The article adeptly elucidates the issue of adverse pregnancy outcomes and their correlation with maternal nutritional adjustments. It underscores the gravity of this problem, given that approximately 10 to 20% of pregnancies are affected by complications.

Significance of Micronutrients: The article offers a comprehensive overview of the multifaceted roles played by micronutrients during pregnancy. It delves into how these micronutrients influence a variety of physiological processes, including hormone secretion, collagen and elastin crosslinking, immune responses, and neurodevelopment.

Overuse of Micronutrient Supplementation: The article correctly identifies the absence of well-defined guidelines as a major concern. It highlights that the excessive or inappropriate utilization of multiple micronutrient supplements during pregnancy has resulted in adverse outcomes, such as spontaneous abortion and gestational diabetes mellitus.

Need for Systematic Studies: The article rightly emphasizes the necessity for systematic research to optimize micronutrient supplementation during each trimester of pregnancy. Such an approach could potentially avert complications associated with both deficiency and excessive supplementation.

Minor Points:

The article could be enhanced by a more extensive discussion of specific micronutrients and their distinct impacts on pregnancy outcomes. This could provide a more nuanced comprehension of the subject matter.

While the article mentions the sources it utilized, a dedicated section detailing the methodology, encompassing the search strategy and inclusion/exclusion criteria for the considered studies, would enhance clarity.

Overall Opinion:

This article constitutes a valuable addition to the field of maternal and fetal health by addressing the critical concern of micronutrient supplementation during pregnancy and its potential consequences on adverse outcomes. The problem is elucidated effectively, and the importance of addressing it is manifest. The article furnishes a comprehensive overview of the roles of micronutrients and appropriately underscores the necessity for systematic studies and standardized guidelines.

To further enrich the article, a more in-depth exploration of individual micronutrients and a more transparent presentation of the research methodology employed for the literature search would be beneficial. In summation, this work is informative and holds promise in guiding future research and clinical practices in maternal nutrition.

Is the topic of the review discussed comprehensively in the context of the current literature?

Yes

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Yes

Are the conclusions drawn appropriate in the context of the current research literature?

Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Biochemistry, endocrinology, obstetrics and gynaecology, nutrition, pharmacology

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard.

Author Response 20 Jan 2024

Krishnananda Prabhu

Dear Sir,

Thank you very much for your valuable input to enrich this article. I shall do the required modification as soon as possible and resubmit for your approval.

Regards,

Competing Interests: No competing interests were disclosed.

Reviewer Report 10 October 2023

<https://doi.org/10.5256/f1000research.137211.r210330>

© 2023 Liu J. This is an open access peer review report distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.



Jufen Liu

¹ Peking University, Beijing, Beijing, China

² Peking University, Beijing, Beijing, China

The review is valuable and of great significance since multiple micronutrients were essential during pregnancy and the intricate relationship between micronutrients and adverse pregnancy outcomes were discussed in this review. For the lack of standardized micronutrient supplementation during pregnancy, the review could guide people to pay attention to optimizing the supplementation of relevant micronutrients and preventing hazards associated with its abuse. The review is clearly structured and easily understood. While the paper addresses a growing subpopulation of importance, the paper has some issues that could be addressed to improve the interpretations:

1. A brief introduction about the common micronutrients and related adverse pregnancy outcome should be given at the first beginning, which would be easily to retrieve the readers' attention.
2. It mentioned that adverse pregnancy complications related to the overuse/unwanted use of multiple micronutrient supplementations during pregnancy in abstract as well as introduction, however, the article is more about micronutrient deficiencies. Consequently, some literature on over-supplementation of micronutrients could be added. And in the part of "Metabolic functions of micronutrients during pregnancy", the status or relevant data of micronutrient supplementations could be mentioned to stress the significance of micronutrient supplementations.
3. In the introduction of various micronutrients in different pregnancy outcomes, the article seems to mainly focus on the role of micronutrients and is lack of research or experiments to prove its point forcefully. This leads to people not being aware of the importance of micronutrient supplementation. And the article shows that there are no proper validation/guidelines available concerning dosing/time, the need for initiation and duration of supplementation, etc. Perhaps some current research in this area could be added to arouse people's thinking.
4. It seems that the conclusion focuses more on the specific factors related mechanism, the

public health significance of appropriate micronutrient supplementation during pregnancy should be added in conclusion.

5. The abstract and conclusion needs to be re-organized and revised.

Is the topic of the review discussed comprehensively in the context of the current literature?

Yes

Are all factual statements correct and adequately supported by citations?

Yes

Is the review written in accessible language?

Yes

Are the conclusions drawn appropriate in the context of the current research literature?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Prenatal nutrition epidemiology

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 10 Oct 2023

Krishnananda Prabhu

Dear Sir,

Thank you very much for your valuable inputs. I shall accordingly include these suggestions and revert soon. This will go a long way in enriching this review article

Regards,

Competing Interests: No competing interests were disclosed.

Author Response 03 Nov 2023

Krishnananda Prabhu

Dear Sir,

Based on your suggestions, I have edited the article. Herewith sending it today to the editorial team for further processing. I sincerely thank for your valuable inputs

Regards,

Competing Interests: I have no conflict of interest whatsoever

The benefits of publishing with F1000Research:

- Your article is published within days, with no editorial bias
- You can publish traditional articles, null/negative results, case reports, data notes and more
- The peer review process is transparent and collaborative
- Your article is indexed in PubMed after passing peer review
- Dedicated customer support at every stage

For pre-submission enquiries, contact research@f1000.com

F1000Research